QUANT

Ratio & Partnership, Mixtures & Alligations

Assignment 02

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Ratio & Mixture Online Assignment -02

1. The ratio of two numbers is 8 : 5. If 8 is added to the greater number and 5 is subtracted from the smaller number, the greater number becomes twice the smaller one. Find the numbers.

1. 72 : 82 2. 71 : 95 3. 72 : 45 4. 70 : 45

2. The ratio of the first and second-class fares between the two stations is 4 : 1 and the number of passengers traveling by first and second-class is 1 : 40. If Rs 1,100 is collected as fare, what is the amount collected from first class passengers?

1. Rs 100 2. Rs 200 3. Rs 150 4. Rs 200

3. 280 coins consists of 1 rupee, 50 paise and 25 paise coins, their values being in the ratio of 20 : 8 : 5. Find the number of coins of each type.

1. 80, 100, 90 2. 80, 90, 100 3.100, 100, 80 4.100, 80, 100

4. The distance of the horizon from a place varies directly as the square root of the height of the place. From a place 100 metres high, the distance of the horizon is 2 km. To which height should one go in order that the distance of the horizon should be 6 km?

1. 300 m 2.640 m 3.540 m 4. 900 m

5. A locomotive engine without any wagon attached to it can run at the maximum speed of 120 km/hr. Its speed is diminished by a quantity, which varies directly as the square root of the number of wagons attached to it. When 16 wagons are attached, its maximum speed is 64 km/hr. Find the greatest number of wagons the engine can carry.

1. 74 2. 73 3.72 4. 75

6. If limestone and gypsum are mixed in the ratio of 5 : 1, then the cement is termed as Ordinary Portland Cement (OPC) and if the same is in the ratio of 7 : 1, then the cement is termed as Portland Poslana Cement (PPC). In a ton of OPC cement, how many kg of limestone need to be added to make it PPC cement?

1.500 kg 2.200 kg 3.333.33 kg 4.166.67 kg

7. The proportion of engineers, diploma holders and other graduates working in a company was 8:5: 1. The salaries of each of them respectively were in the proportion of 5:2: 3. The company employed 20 diploma holders and the total daily salary for all employees was Rs 318. The daily salary paid to an engineer was

1.Rs 10 2.Rs 7.5 3.Rs 15 4.None of the above

8. What proportion of a mixture containing alcohol and water in the proportion of 7:5 should be removed and equivalent quantity of water added so that the resulting mixture has both the liquids in equal proportion?

1. $^{2}/_{7}$ 2. $^{1}/_{7}$ 3. $^{5}/_{12}$ 4. $^{12}/_{7}$

9. A merchant was selling his goods at 20 % profit. When he allowed a discount of 5 paise per rupee on sale, his sale improved in the ratio of 8 : 5. His new profit / old profit ratio is

1.28 : 25 2.22 : 25 3.30 : 26 4.38 : 25

10. Two liquids are mixed in the ratio of 5: 7 and sold at a profit of 25% at Rs 10 a liter. If the first liquid costs Rs 3 more than the second one, what was the cost price of both the liquids?

1.9.25: 6.25 2.11: 8 3.9.75: 6.75 4.10.5: 7.5

Answer key & Explanations

Q No.	Key	Explanation
1.	3	Present ratio = 8 : 5. $\therefore$ actual values are $8x$ and $5x$. Hence $8x + 8 : 5x - 5 = 2 : 1$ $\Rightarrow x = 9$. $\therefore$ numbers are 8(9) = 72 and 5(9) = 45. So the ratio is 72: 45.
2.	1	Ratio of the amounts collected from 1st and 2nd class = (4 × 1): (1 × 40) =1 : 10. $\therefore$ Amount collected from 1st class passengers = $\frac{1}{11} \times 1100$ = 100.
3.	4	Value of rupee coins = Rs 20 i.e. 20 coins. Value of 50 p coins = Rs 8 i.e. 16 coins. Value of 25 p coins = Rs 5 i.e. 20 coins. $\therefore$ Ratio of coins = 20 : 16 : 20 $\Rightarrow$ 5 : 4 : 5. $\therefore$ Number of rupee coins =5/14 × 280 =100. Number of 50 *P* coins = 4/14 × 280 = 80 and Number of 25 *P* coins = 5/14 × 280 = 100.
4.	4	D $\alpha \sqrt{H}.D = k\sqrt{H}.$ Also 2 = $k\sqrt{100} \Rightarrow k = 2/10 = 1/5.$ Now 6 = $\frac{1}{5}\sqrt{H}$ $\Rightarrow \sqrt{H} = 30$ $\Rightarrow H = 900$ m.
5.	2	Max speed = 120 Km/hr. When 16 wagons are attached, maximum speed = 64 km/hr. $\therefore 120 - 64 \propto \sqrt{16} \Rightarrow 56 = k\sqrt{16} \Rightarrow k = 14.$ Greatest number of wagons the engine can carry will be 120 = 14 $\sqrt{w}$ $\Rightarrow w = 73.46.$ So the answer is 73. Because anything above this value will stop the train.
6.	3	1 Ton = 1000 Kg. Let x kg of limestone be added to make PPC from *OPC*. $\therefore (\frac{5000}{6} + x)/(1000 + x) = \frac{7}{8}.$ $\Rightarrow x = 333.33$ kg.
7.	2	$E : DH : G$ = 8 : 5 : 1. There are 20 diploma holders. $\therefore$ there are 32 engineers and 4 graduates. Their salary ratio = 5 : 2 : 3 $\Rightarrow x : y : z$ = 5 : 2 : 3. Total daily salary for all employees = 318. $\Rightarrow 32x + 20y + 4z = 318$ $\Rightarrow 32x + 20\left(\frac{2}{5}x\right) + 4\left(\frac{3}{5}x\right)$ $= 318 \Rightarrow x = 7.5.$
8.	2	Both the liquids in equal proportion. $\therefore$ proportion is ½. Apply the relation: $\frac{\text{Amount of } A \text{ left}}{\text{Amount of } A \text{ originally present}} = \left(1 - \frac{R}{M}\right)^n$ Present ratio = 7 : 5. $\therefore \frac{1/2}{7/12} = \left(1 - \frac{R}{1}\right)^1$ $\Rightarrow$ R =1/7.

9.	1	Let the CP be Re. 1, profit will be Rs 0.20 and SP will be 1.20. Now the new selling price is $(1.20 \times 95)/100 = 1.14$. Now his sale has become 8/5 times of the original i.e. $1.14 \times 8/5 = 1.824$. Now the new cost price will be 8/5 times of the original CP because he is selling more quantity. $100 \times 8/5$ = Rs 160. His new profit is $1.824 - 160 = 0.224$. Ratio of new profit to old profit = 0.224 : 0.200 $\Rightarrow$224 : 200 $\Rightarrow$ 28 : 25.
10.	3	Ratio = 5 : 7. SP = 10. Profit = 25%. $\therefore$CP = 8. Let the 1st liquid costs $x - 3$ $\therefore$cost of 2nd liquid = x. Hence $x - 8$: $8 - (x - 3) = 7 : 5 \Rightarrow x = 9.75$. $\therefore$Cost of 2nd liquid = 6.75.